

Network Incident Investigation Report

Dynamic RCA & Timeline Reconstruction Analysis

Total Reconstructed Incidents:	13	Total Analyzed Log Events:	45
Investigation Date:	2026-06-29	Analyzed Log Sources:	Access-6100-01, Access-6100-02, Access-6100-03, Access-6100-04, Core-8325-01, Dist-6300-01

Incident Overview Table

Incident ID	Start Time	End Time	Duration	Event Count	Affected Devices
WORKFLO W-0001	2026-09-01T10:00:01	2026-09-01T10:04:55	4.9m	4	Core-8325-01
WORKFLO W-0002	2026-09-01T10:00:05	2026-09-01T10:04:30	4.4m	3	Core-8325-01
WORKFLO W-0003	2026-09-01T10:00:08	2026-09-01T10:04:40	4.5m	2	Access-6100-01
INC-0004	2026-09-01T10:01:00	2026-09-01T10:04:10	3.2m	16	Dist-6300-01
INC-0005	2026-09-01T10:01:22	2026-09-01T10:01:26	4.0s	2	Core-8325-01
INC-0006	2026-09-01T10:02:00	2026-09-01T10:02:06	6.0s	2	Access-6100-02
INC-0007	2026-09-01T10:02:15	2026-09-01T10:03:20	1.1m	7	Core-8325-01
INC-0008	2026-09-01T10:03:00	2026-09-01T10:03:10	10.0s	4	Access-6100-03
WORKFLO W-0009	2026-09-01T10:03:15	2026-09-01T10:03:15	0.0s	1	Core-8325-01
WORKFLO W-0010	2026-09-01T10:04:20	2026-09-01T10:04:20	0.0s	1	Core-8325-01
WORKFLO W-0011	2026-09-01T10:04:45	2026-09-01T10:04:45	0.0s	1	Access-6100-01
WORKFLO W-0012	2026-09-01T10:05:00	2026-09-01T10:05:00	0.0s	1	Access-6100-04
WORKFLO W-0013	2026-09-01T10:05:05	2026-09-01T10:05:05	0.0s	1	Access-6100-04

Severity Matrix

Severity Level	Event Count	% of Total
Critical	3	6.7%
Warning / Error	10	22.2%
Info	32	71.1%
Total	45	100%

Network Domain Classification

Domain Tier	Event Count	% of Total
Layer 1 - Physical	11	24.4%
Layer 2 - Switching	5	11.1%
Layer 4 - Management	5	11.1%
Layer 4 - Security	1	2.2%
Unknown / Other	23	51.1%

Detailed Investigation Report

Network Incident Investigation Report

Generated: 2026-06-29T13:52:38Z

Executive Summary

- Total incidents reconstructed: 5
- Total events analyzed: 45
- Total causal links inferred: 36
- Affected devices: Access-6100-02, Access-6100-03, Core-8325-01, Dist-6300-01

Probable Initiating Triggers

- Incident INC-0004 -> transceiver (device=Dist-6300-01, score=192.2)
- Incident INC-0006 -> dot1x_failure (device=Access-6100-02, score=54.3)
- Incident INC-0007 -> power_failure (device=Core-8325-01, score=218.8)
- Incident INC-0008 -> crc_errors (device=Access-6100-03, score=188.9)

Incident Overview

- INC-0004: events=16, duration=190.0s, primary_issue=transceiver
- INC-0005: events=2, duration=4.0s, primary_issue=ssh_bruteforce
- INC-0006: events=2, duration=6.0s, primary_issue=dot1x_failure
- INC-0007: events=7, duration=65.0s, primary_issue=power_failure
- INC-0008: events=4, duration=10.0s, primary_issue=crc_errors

Detailed Incident Chains

INC-0004

Failure Sequence:

- transceiver (info)
- interface_down (info)
- interface_down (info)
- ospf_interface_down (info)
- ospf_neighbor_down (warning)
- route_recalculation_started (info)
- route_recalculation_completed (info)
- bgp_session_lost (warning)
- route_withdrawal (info)

Recovery Sequence:

- transceiver (info)
- interface_up (info)
- interface_up (info)
- ospf_interface_up (info)
- ospf_neighbor_up (info)
- bgp_session_established (info)
- routes_relearned (info)

Status: Resolved

Duration: 190.0s

INC-0005

Failure Sequence:

- ssh_bruteforce ×3 (warning)
- ssh_source_blocked (critical)

Status: Active

Duration: 4.0s

INC-0006

Failure Sequence:

- dot1x_failure ×3 (error)
- port_blocked (warning)

Status: Active

Duration: 6.0s

INC-0007

Failure Sequence:

- power_failure (critical)
- fan_failure (warning)
- thermal (warning)
- thermal (warning)
- linecard_disabled (critical)
- bgp_session_lost (warning)
- lldp (info)

Status: Active

Duration: 65.0s

INC-0008

Failure Sequence:

- crc_errors x2 (warning)
- interface_down (info)
- interface_down (info)
- lldp (info)

Status: Active

Duration: 10.0s

Operational Workflows Detected

Workflow: WORKFLOW-0001

- **Status:** Successful
- **Incident Detected:** No
- **Sequence Summary:**
 - ntp (info)
 - snmp (info)
 - ntp (info)
 - snmp (info)

Workflow: WORKFLOW-0002

- **Status:** Successful
- **Incident Detected:** No
- **Sequence Summary:**
 - config_change (info)
 - raw (info) x2

Workflow: WORKFLOW-0003

- **Status:** Successful
- **Incident Detected:** No
- **Sequence Summary:**
 - mac_auth (info) x2

Workflow: WORKFLOW-0010

- **Status:** Successful
- **Incident Detected:** No

- **Sequence Summary:**

- raw (info)

Routine & Unlinked Noise

The following events were classified as non-actionable noise or routine informational activity:

- Core-8325-01: vlan (info) - VLAN 200 created
- Access-6100-01: lldp (info) - LLDP neighbor discovered on port 1/1/12
- Access-6100-04: interface_up (info) - Port 1/1/20 is now on-line
- Access-6100-04: interface_up (info) - Port 1/1/21 is now on-line

Confidence and Limitations

- Causality is inferred from temporal and contextual heuristics, not strict proof.
- Confidence increases when links have strong timing, device/interface alignment, and severity progression.

Recommendations

- Prioritize remediation on root-linked interfaces/devices before downstream symptoms.
- Add monitoring alerts for repeated trigger subtypes and interface recurrence.
- Validate inferred root causes with device-level diagnostics and config audit.